B.TECH PROJECT PRESENTATION • 2026–2027

Warehouse Management Portal

A Web-Based System for Inventory and Warehouse Operations

SUBMITTED BY

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OVERVIEW

Introduction

Web-Based System

Warehouse Management Portal is a complete web-based application designed to streamline facility operations.

Core Management

Helps manage products, stock levels, suppliers, and customer orders from a single interface.

Efficiency & Accuracy

Reduces manual work, keeps information organized, and improves stock tracking accuracy.

CHALLENGES

Problem Statement

- **Manual Stock Data**
  Stock data maintained manually leads to high error rates.
- **Information Discrepancies**
  Product information can be lost or entered incorrectly.
- **Low-Stock Identification**
  Difficult to timely identify low stock items before stockouts occur.
- **Unorganized Records**
  Supplier and order data remain fragmented and unorganized.
- **Time-Consuming Workflow**
  Manual work consumes excessive time and delays dispatch.
- **Tracking Complexity**
  Tracking overall inventory becomes difficult as operations scale.

SOLUTION

Proposed Solution

Digital Warehouse Control

The Warehouse Management Portal provides a centralized digital solution to manage all core warehouse activities efficiently.

KEY CAPABILITIES

- ✓ Add and manage products seamlessly
- ✓ Monitor inventory and track stock quantity
- ✓ Maintain comprehensive supplier details
- ✓ Create and manage customer orders
- ✓ Instant low-stock alerts & dashboard analytics

GOALS

Project Objectives

1. **System Development**
   Develop a simple, reliable warehouse management system.
2. **Digital Cataloging**
   Manage product information digitally with structured records.
3. **Inventory Tracking**
   Track inventory and available stock in real time.
4. **Supplier Details**
   Maintain comprehensive and updated supplier records.
5. **Order Efficiency**
   Manage customer orders efficiently to avoid fulfillment delays.
6. **Error Reduction**
   Reduce manual errors and improve speed & accuracy in stock management.

ARCHITECTURE

Main Modules (1 / 2)

1. Dashboard

Shows overall products, stock levels, suppliers, and customer orders at a glance.

2. Product Management

Add, view, edit, and delete product details and catalog parameters easily.

3. Inventory Management

Monitor stock quantity and update available inventory dynamically.

ARCHITECTURE

Main Modules (2 / 2)

4. Supplier Management

Store supplier name, contact info, and company details in one organized directory.

5. Order Management

Create, update, and track customer orders smoothly through fulfillment.

6. Low Stock Alert

Automated warnings to help identify products that need immediate restocking.

TECH STACK

Technologies Used

HTML5
Structure of web pages.

CSS3
Design and styling.

JavaScript
Interactivity & dynamics.

Bootstrap
Responsive interface.

Local Storage
Client data persistence.

PROCESS FLOW

System Workflow

01
User opens the application

02
User views the dashboard

03
User handles products

04
User updates stock quantity

05
User maintains suppliers

06
User creates/tracks orders

SUMMARY

Conclusion

- Provides a simple digital solution for warehouse operations.
- Manages products, stock, suppliers, and orders in one place.
- Removes manual effort and significantly improves data accuracy.
- Makes inventory monitoring faster and more organized.

Future Scope

The project can be enhanced in future versions with advanced features and AI-based predictive stock forecasting.

ROADMAP

Future Enhancements

Barcode & QR Scanning

Integration of optical scanners for rapid item check-in and check-out.

AI Demand Forecasting

Machine learning models to predict stock requirements based on trends.

Cloud Deployment

Scaling the web portal for multi-warehouse remote management.

Thank You

Questions & Discussion

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